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ASSIGNMENT 1

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples

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Frontend development is the part of a website that users can see and interact with. It focuses on the design, layout, buttons, menus, and overall user experience. Frontend developers mainly use HTML, CSS, and JavaScript to build attractive and responsive web pages.

Backend Development

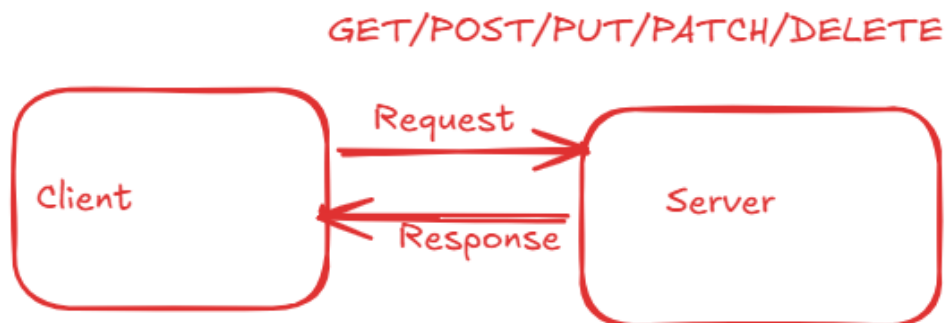
Backend development is the part that works behind the scenes. It is responsible for managing the server, database, and application logic. The backend processes user requests, stores data securely, and ensures that the website functions correctly.

Full-Stack Development

Full-stack development includes both frontend and backend development. A full-stack developer has the knowledge and skills to build and manage both the user interface and the server-side of a web application.

2. Create a simple diagram showing how the client-server model works in web architecture.

CLIENT/SERVER Model--



3. Describe how a browser requests and displays a web page from a web server.

ANS When a user enters a website address (URL) in a web browser and presses **Enter**, the browser sends an **HTTP request** to the web server. The server receives the request, processes it, and finds the required web page or data. It then sends an **HTTP response** back to the browser, which usually contains HTML, CSS, JavaScript, images, and other resources.

After receiving the response, the browser reads the HTML to understand the structure of the page, applies CSS for styling, and executes JavaScript to make the page interactive. Finally, the browser combines all these elements and displays the complete web page on the user's screen.

Steps Involved

1. The user enters a URL in the browser.
2. The browser sends an HTTP request to the web server.
3. The web server processes the request and retrieves the required files.
4. The server sends an HTTP response back to the browser.
5. The browser renders the HTML, CSS, and JavaScript.
6. The complete web page is displayed to the user.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

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Code Editor (VS Code, Sublime Text)	Used to write and edit HTML, CSS, JavaScript, and other programming code.
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Web Browser (Google Chrome, Mozilla Firefox)	Used to open, test, and view web pages during development.
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Browser Developer Tools	Help inspect web pages, debug errors, and test the website's layout and performance.
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Web Server (XAMPP, WAMP, Node.js)	Runs the website locally and processes server-side code during development.
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Version Control (Git)	Tracks changes in the project and helps manage different versions of the code.
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GitHub	Stores project code online and makes it easy to collaborate with others.
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5. Explain what a web server is and give examples of commonly used servers.

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A web server is a software or system that stores, processes, and delivers web pages to users through the internet. When a user requests a website using a web browser, the web server receives the request, finds the required files, and sends them back to the browser. It helps websites run smoothly by handling requests and serving web content.

Commonly Used Web Servers

- Apache HTTP Server – One of the most popular open-source web servers. It is reliable, flexible, and widely used for hosting websites.
- Nginx – A high-performance web server known for its speed and ability to handle many users at the same time.
- Microsoft Internet Information Services (IIS) – A web server developed by Microsoft for hosting websites and applications on Windows.
- LiteSpeed Web Server – A fast and secure web server that improves website performance and is commonly used for high-traffic websites.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

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Frontend Developer

A frontend developer is responsible for creating the part of the website that users see and interact with. They design the layout, improve the user experience, and make sure the website is responsive and easy to use. They mainly work with HTML, CSS, and JavaScript.

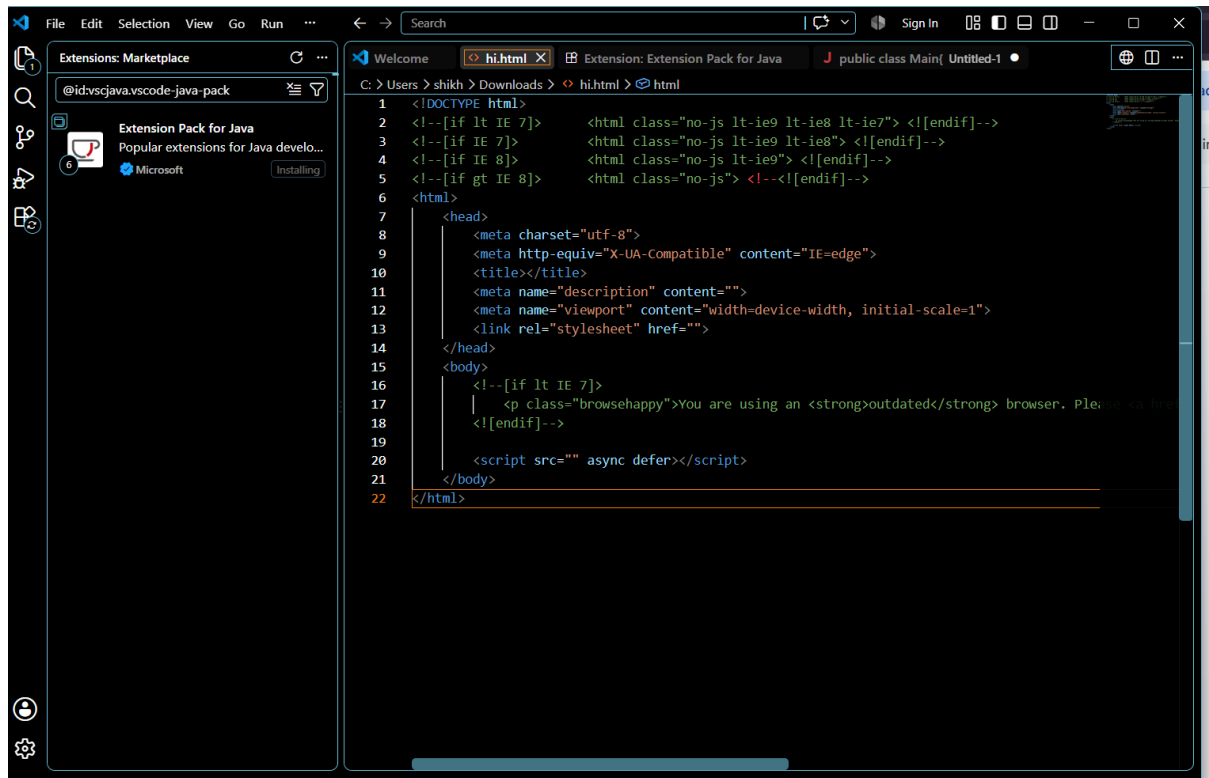
Backend Developer

A backend developer works on the server side of the application. They develop the business logic, manage user requests, connect the application to the database, and ensure that the website functions correctly. They commonly use languages such as Java, Python, PHP, or Node.js.

Database Administrator (DBA)

A Database Administrator (DBA) is responsible for managing and maintaining the project's database. They organize data, ensure data security, perform backups, and optimize database performance so that information can be stored and retrieved efficiently.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.



8. Explain the difference between static and dynamic websites. Provide an example of each.

Static Website

A static website displays the same content to every user. The web pages are created using HTML, CSS, and sometimes JavaScript, and the content does not change unless the developer updates the files manually. Static websites are simple, fast, and suitable for websites with fixed information.

Example: A personal portfolio website or a company's basic information website.

Dynamic Website

A dynamic website displays different content based on user actions or data stored in a database. It uses server-side programming languages such as PHP, Python, Java, or Node.js along with databases like MySQL. Dynamic websites are interactive and can update content automatically.

9. Research and list five web browsers. Explain how rendering engines differ between them.

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- **Blink** is used by Google Chrome, Microsoft Edge, and Opera. It is known for its fast performance, efficient rendering, and support for modern web technologies.
- **Gecko** is used by Mozilla Firefox. It focuses on web standards, privacy, and independent browser development.
- **WebKit** is used by Apple Safari. It is optimized for Apple devices and provides good speed, power efficiency, and smooth performance.

Although all rendering engines display web pages, they may process HTML, CSS, and JavaScript slightly differently. This can sometimes cause a website to look or behave differently in different

browsers. Therefore, web developers test their websites in multiple browsers to ensure they work correctly for all users.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Text Diagram:

